

Business Analytics

Data-driven insights for strategic and operational decision-making.

Programme level	Graduate, Postgraduate	Duration	12 weeks
Credits	3	Delivery	Classroom, Online, Hybrid
Subject area	Business Analytics	Updated	2026-07-21

Course overview

A practice-oriented course that develops the ability to frame business questions, analyse data, interpret models and communicate evidence for managerial action.

Intended audience: Business management, engineering management and early-career professionals who need practical analytical decision skills.

Prerequisites: Basic spreadsheet skills; no advanced programming required.

Learning outcomes

- Frame business problems as answerable analytical questions.
- Prepare and examine structured business data.
- Select suitable descriptive, predictive and prescriptive approaches.
- Interpret analytical results for managerial decisions.
- Communicate recommendations through clear visual and written narratives.

Module structure

01	Business questions and analytical thinking Problem framing, decision context and measures of success.
02	Data preparation and exploration Data quality, summaries, patterns and visual analysis.
03	Predictive analysis for management Forecasting, classification and practical model interpretation.
04	Decision models and scenarios What-if analysis, optimisation and uncertainty.
05	Communicating analytical recommendations Dashboards, executive narratives and implementation choices.

Assessment pattern

Assessment component	Weight
Applied analysis exercises	30%
Business case report	30%
Final analytics project	40%

Delivery note

The module sequence, class activities and assessment details may be adapted for programme schedules, cohort experience and institutional requirements. The course remains focused on practical application, structured judgement and professional communication.